

Chapter 2 – Wazuh Architecture

Learning Objectives

After completing this chapter, you will be able to:

- Understand the architecture of Wazuh.
 - Identify the role of each Wazuh component.
 - Understand how data flows through the platform.
 - Learn how agents communicate with the Wazuh Manager.
 - Understand the purpose of the Wazuh Indexer and Dashboard.
 - Prepare for the installation process in the following chapters.
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2.1 What is Wazuh Architecture?

Before installing Wazuh, it is important to understand **how its components work together**.

Think of Wazuh as a security monitoring system made up of several specialized services. Each service performs a different task, and together they provide centralized visibility into your infrastructure.

Understanding the architecture helps you:

- Troubleshoot issues more effectively.
 - Understand how alerts are generated.
 - Deploy agents correctly.
 - Expand your environment in the future.
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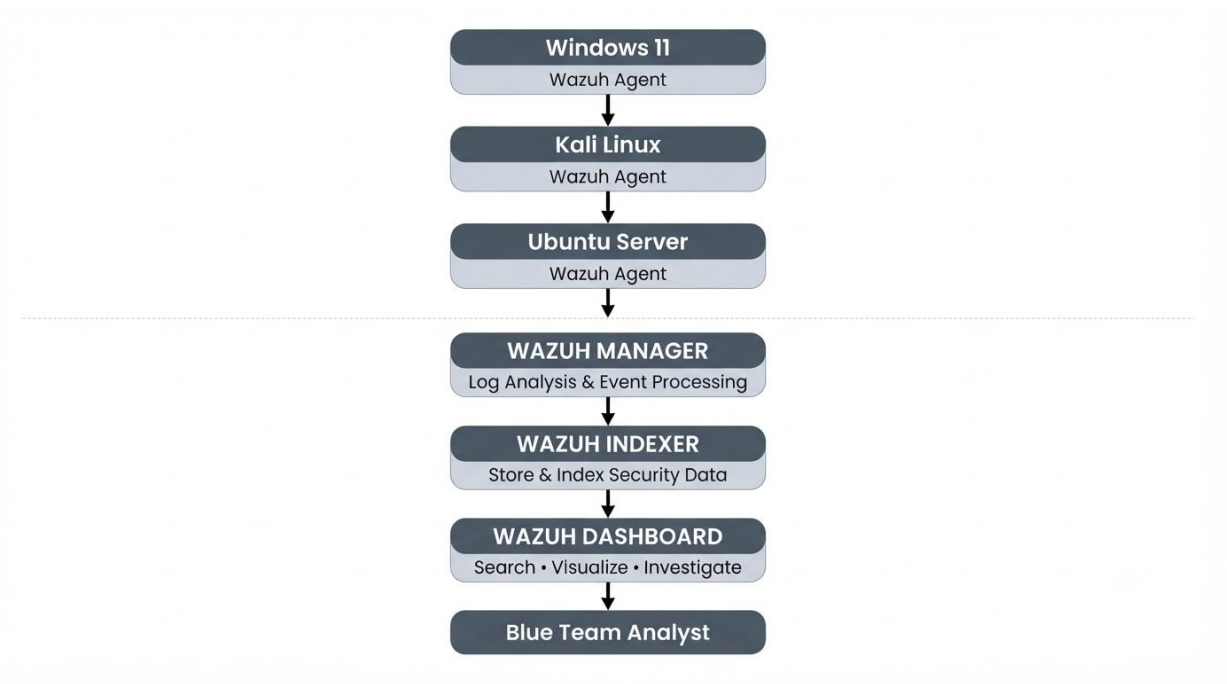
2.2 Wazuh Architecture Overview

A standard Wazuh deployment consists of four main components:

1. Wazuh Manager
2. Wazuh Indexer
3. Wazuh Dashboard
4. Wazuh Agents

These components work together to collect, process, store, and display security information.

Wazuh Architecture Diagram



2.3 Wazuh Manager

The **Wazuh Manager** is the core component of the Wazuh platform.

It is responsible for:

- Receiving data from agents.
- Analyzing security events.
- Detecting suspicious activity.

- Applying security rules.
- Generating alerts.
- Managing connected agents.

Think of the Manager as the **brain** of Wazuh.

Without the Manager, the other components cannot perform meaningful security analysis.

Responsibilities of the Wazuh Manager

- Log Analysis
- Rule Matching
- Event Correlation
- Agent Management
- Active Response
- Alert Generation

Manager Workflow

Agent Sends Log→Receive Event→Analyze Log→Apply Detection
Rule→Generate Alert→Send to Indexer

```

kashif@wazuh-server:~$ sudo systemctl restart wazuh-manager
kashif@wazuh-server:~$ sudo systemctl status wazuh-manager
• wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-07-21 12:52:17 UTC; 2s ago
     Process: 2192 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/0)
    Tasks: 152 (limit: 5776)
      Memory: 816.4M
         CPU: 17.718s
    CGroup: /system.slice/wazuh-manager.service
           └─1100 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              1101 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              1102 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              1105 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              1108 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              1153 /var/ossec/bin/wazuh-authd
              1174 /var/ossec/bin/wazuh-db
              1190 /var/ossec/bin/wazuh-execd
              1208 /var/ossec/bin/wazuh-analysisd
              1235 /var/ossec/bin/wazuh-syscheckd
              1255 /var/ossec/bin/wazuh-remoted
              1304 /var/ossec/bin/wazuh-logcollector
              1326 /var/ossec/bin/wazuh-monitord
              2314 /var/ossec/bin/wazuh-modulesd
              2872 /usr/bin/python3 /usr/sbin/ufw status

Jul 21 12:52:13 wazuh-server env[2192]: wazuh-syscheckd already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-remoted already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-logcollector already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-monitord already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-modulesd: Process 62068 not used by Wazuh, removing...
Jul 21 12:52:13 wazuh-server env[2311]: 2026/07/21 12:52:13 wazuh-modulesd:router: INFO: Loaded rou
Jul 21 12:52:13 wazuh-server env[2311]: 2026/07/21 12:52:13 wazuh-modulesd:content_manager: INFO: L
Jul 21 12:52:14 wazuh-server env[2192]: Started wazuh-modulesd...
Jul 21 12:52:17 wazuh-server env[2192]: Completed.
Jul 21 12:52:17 wazuh-server systemd[1]: Started Wazuh manager.
lines 1-34/34 (END)

```

2.4 Wazuh Indexer

The **Wazuh Indexer** stores all security events processed by the Manager.

It is built on **OpenSearch**, which is optimized for storing and searching large amounts of data.

Its responsibilities include:

- Storing alerts.
- Indexing logs.
- Fast searching.
- Data retrieval.
- Supporting dashboard queries.

Think of the Indexer as the **database** of Wazuh.

Indexer Workflow

Manager→Receive Alerts→Index Data→Store Data→Dashboard Searches

```
kashif@wazuh-server:~$ sudo systemctl status wazuh-manager wazuh-indexer wazuh-dashboard filebeat --no-pager
[sudo] password for kashif:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-07-21 07:41:32 UTC; 2min 50s ago
     Tasks: 154 (limit: 5776)
    Memory: 1.0G
       CPU: 1min 21.440s
    CGroup: /system.slice/wazuh-manager.service
            └─61818 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
              └─61819 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                └─61821 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                  └─61825 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                    └─61829 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                      └─61868 /var/ossec/bin/wazuh-authd
                        └─61884 /var/ossec/bin/wazuh-db
                          └─61900 /var/ossec/bin/wazuh-execd
                            └─61926 /var/ossec/bin/wazuh-analysisd
                              └─61939 /var/ossec/bin/wazuh-syscheckd
                                └─61985 /var/ossec/bin/wazuh-remoted
                                  └─62019 /var/ossec/bin/wazuh-logcollector
                                    └─62042 /var/ossec/bin/wazuh-monitord
                                      └─62068 /var/ossec/bin/wazuh-modulesd

Jul 21 07:41:24 wazuh-server env[61750]: Started wazuh-analysisd...
Jul 21 07:41:25 wazuh-server env[61750]: Started wazuh-syscheckd...
Jul 21 07:41:26 wazuh-server env[61750]: Started wazuh-remoted...
Jul 21 07:41:27 wazuh-server env[61750]: Started wazuh-logcollector...
Jul 21 07:41:28 wazuh-server env[61750]: Started wazuh-monitord...
Jul 21 07:41:28 wazuh-server env[62065]: 2026/07/21 07:41:28 wazuh-modulesd:router: INFO: Loaded router module.
Jul 21 07:41:28 wazuh-server env[62065]: 2026/07/21 07:41:28 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Jul 21 07:41:30 wazuh-server env[61750]: Started wazuh-modulesd...
Jul 21 07:41:32 wazuh-server env[61750]: Completed.
Jul 21 07:41:32 wazuh-server systemd[1]: Started Wazuh manager.

● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/lib/systemd/system/wazuh-indexer.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-07-21 07:04:18 UTC; 40min ago
     Docs: https://documentation.wazuh.com
    Main PID: 8453 (java)
      Tasks: 67 (limit: 5776)
     Memory: 1.3G
        CPU: 3min 22.293s
```

2.5 Wazuh Dashboard

The **Wazuh Dashboard** is the graphical web interface used by security analysts.

Instead of reading log files manually, analysts use the Dashboard to:

- Search logs.
- View alerts.
- Monitor agents.
- Create visualizations.
- Investigate incidents.
- Review vulnerabilities.

Think of the Dashboard as the **control center** for your security operations.

Dashboard Features

- Security Dashboard
- Alert Monitoring
- Agent Management
- Threat Detection
- Vulnerability Reports
- Compliance Reports
- Log Search
- Event Visualization

```

└─1304 /var/ossec/bin/wazuh-logcollector
└─1326 /var/ossec/bin/wazuh-monitord
└─2314 /var/ossec/bin/wazuh-modulesd
└─2872 /usr/bin/python3 /usr/sbin/ufw status

Jul 21 12:52:13 wazuh-server env[2192]: wazuh-syscheckd already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-remoted already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-logcollector already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-monitord already running...
Jul 21 12:52:13 wazuh-server env[2192]: wazuh-modulesd: Process 62068 not used by Wazuh, removing...
Jul 21 12:52:13 wazuh-server env[2311]: 2026/07/21 12:52:13 wazuh-modulesd:router: INFO: Loaded rou>
Jul 21 12:52:13 wazuh-server env[2311]: 2026/07/21 12:52:13 wazuh-modulesd:content_manager: INFO: L>
Jul 21 12:52:14 wazuh-server env[2192]: Started wazuh-modulesd...
Jul 21 12:52:17 wazuh-server env[2192]: Completed.
Jul 21 12:52:17 wazuh-server systemd[1]: Started Wazuh manager.
kashif@wazuh-server:~$ sudo systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-07-21 12:26:11 UTC; 27min ago
     Main PID: 684 (node)
        Tasks: 11 (limit: 5776)
       Memory: 310.0M
          CPU: 1min 44.016s
      CGroup: /system.slice/wazuh-dashboard.service
              └─684 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=65>

Jul 21 12:53:31 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:33 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:36 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:38 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:41 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:43 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:46 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:48 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:51 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
Jul 21 12:53:53 wazuh-server opensearch-dashboards[684]: {"type":"log","@timestamp":"2026-07-21T12:>
lines 1-20/20 (END)
```

2.6 Wazuh Agent

The **Wazuh Agent** is installed on systems you want to monitor.

Examples include:

- Windows
- Linux
- macOS

The Agent continuously collects information such as:

- Authentication logs
- File changes
- Running processes
- System information
- Installed software
- Security events

The collected data is securely sent to the Wazuh Manager.

2.7 Data Flow in Wazuh

Understanding the flow of security data is one of the most important concepts in Wazuh.

Complete Data Flow



Every security event follows this path before it appears on the Dashboard.

2.8 How Alerts are Generated

Wazuh does not generate alerts randomly.

The process follows these steps:

1. A system generates a log.
 2. The Agent collects the log.
 3. The Manager receives the log.
 4. Detection rules analyze the log.
 5. If a rule matches, an alert is created.
 6. The alert is stored in the Indexer.
 7. The Dashboard displays the alert.
-

2.9 Communication Between Components

Each component communicates securely with the others.

Source	Destination	Purpose
Agent	Manager	Send logs
Manager	Indexer	Store alerts
Dashboard	Indexer	Query data
Analyst	Dashboard	View alerts

Communication Flow

Agent→Manager→Indexer→Dashboard→Security Analyst

2.10 Wazuh Deployment Types

Wazuh supports multiple deployment models.

Single Node

All components run on one server.

Suitable for:

- Home Labs
 - Students
 - Small Organizations
-

Distributed Deployment

Components run on separate servers.

Suitable for:

- Medium Businesses
 - Large Enterprises
 - High Availability
-

Single Node Diagram

Ubuntu Server

|— Wazuh Manager

|— Wazuh Indexer

|— Wazuh Dashboard

Distributed Deployment

Server 1

Manager

|

Server 2

Indexer

|

Server 3

Dashboard

2.11 Ports Used by Wazuh

Port	Service	Purpose
443	Dashboard	Secure Web Access
1514	Manager	Agent Communication
1515	Manager	Agent Registration
55000	API	REST API

These ports must be accessible for proper communication between Wazuh components.

2.12 Why Understanding the Architecture Matters

Many beginners install Wazuh without understanding how the components interact.

Knowing the architecture helps you:

- Troubleshoot communication problems.
 - Identify which service has failed.
 - Understand alert generation.
 - Expand your deployment.
 - Configure agents correctly.
-

Best Practices

- Learn the architecture before installation.
 - Use a Single Node deployment for home labs.
 - Keep the Manager, Indexer, and Dashboard on the same server while learning.
 - Ensure agents communicate only with trusted Managers.
 - Document your network topology and IP addresses.
-

Chapter Summary

In this chapter, you learned:

- The four core components of Wazuh.
- The responsibilities of the Manager, Indexer, Dashboard, and Agents.
- How security data flows through the platform.
- How alerts are generated.
- The communication between Wazuh components.
- The difference between Single Node and Distributed deployments.
- The ports used by Wazuh.
- The architecture of the Blue Team home lab.

Understanding this architecture provides the foundation for successfully deploying and managing Wazuh.

One-Line Summary

Wazuh uses a modular architecture in which Agents collect security data, the Manager analyzes it, the Indexer stores it, and the Dashboard presents it to security analysts for monitoring and investigation.
